

R&D engineer in computational imaging, with a background in electrical engineering and medical image analysis. I build tools that let microscopes follow living samples on their own, and pipelines that turn large time-lapse datasets into something biologists can actually use. Before that, I spent four years on 4D MRI of the thorax, trying to help radiologists spot mesothelioma earlier.

CONTACT
INFORMATION

Computational Imaging Platform
Chan Zuckerberg Biohub, San Francisco, CA

✉ taylla.theodoro@czbiohub.org
 [taylla-theodoro](https://www.linkedin.com/in/taylla-theodoro)  [tayllatheodoro](https://github.com/tayllatheodoro)

RESEARCH AND
WORK
EXPERIENCE

Computational Imaging - Chan Zuckerberg Biohub SF Aug 2024 – present
R&D Engineer Microscope automation and high-throughput analysis of live-cell imaging.

- Built DynaTrack, which lets the microscope follow developing zebrafish neuromasts on its own. It reconstructs phase and virtually stains nuclei during acquisition, so tracking needs only gentle bright-field light. It kept 73 of 75 neuromasts in view for up to 70.5 h; before, they drifted out of view after about 12 h [1].
- Curated the live-cell dataset shared by DynaCLR and DynaCell: registered the volumes, segmented nuclei from virtual stains, tracked cells with Ultrack, and organized everything as OME-Zarr. Thousands of infected and drug-treated cells over 3D time-lapses. DynaCLR's infection classifier (F1 98.4) was trained on it [2–4].
- Curated and registered the DynaCell benchmark for dynamic 3D virtual staining: paired label-free and fluorescence volumes of A549 cells, four organelles, three infection conditions, imaged on the Mantis microscope [5].

LIDS - Laboratory of Image Data Science, UNICAMP Aug 2021 – 2025
Graduate Researcher Advisor: Prof. Alexandre Xavier Falcão

- Worked on finding Early Contrast Enhancement, an MRI sign of pleural mesothelioma, automatically in 4D (3D+t) DCE-MRI of the thorax.
- Built Meso-Regions, an open-source pipeline that flags suspicious pleural regions with no manual ROI drawing: ANTs motion correction, pleural extraction with the image foresting transform, SLIC/DISF supervoxels, enhancement curves per supervoxel, and a rule-based detector. Presented at IEEE ISBI 2026 [6, 7].
- Trained a PyTorch U-Net, with diffusion-based augmentation, to segment the pleural region that contains the tumor. Co-authored a superpixel tutorial at SIBGRAPI and the iMig imaging review [8, 9].

PREDICT-Meso Network, University of Glasgow Feb 2023 – Jul 2023
Visiting Researcher Built MISE, an interactive tool for segmenting 4D DCE-MRI, and used it to label the pleura across the PREDICT-Meso cohort [10].

Peapply & Co. May 2021 – Oct 2021
Data Engineer Built data pipelines in PySpark and SQL for collecting, storing, and analyzing data at scale.

LIII - Lab of Image, Intelligence and Innovation Jan 2020 – Apr 2020
Visiting Student Researcher First deep learning project: CNN-based spine segmentation in MRI with TensorFlow.

LAB2I - Automation and Intelligent Instrumentation Laboratory Aug 2017 – Dec 2019
Undergraduate Research Intern Built an embedded system with sensors to monitor temperature and current in electrocoagulation water treatment [11, 12].

EDUCATION

University of Campinas (UNICAMP) SP, BR Aug 2021 – 2025
M.Sc. in Computer Science, GPA 4.0/4.0

- Thesis: Towards Automatic Detection of Pleural Mesothelioma Biomarker in 4D Dynamic MR Imaging [7]
- Advisor: Prof. Alexandre Xavier Falcão

State University of Londrina (UEL) PR, BR Apr 2016 – Mar 2021
B.Sc. in Electrical Engineering

- Thesis: Study of Deep Learning and its Application to Medical Images
- Advisor: Prof. Francisco Granziera Junior

ADVANCED
TRAINING

Marine Biological Laboratory (MBL), Woods Hole, MA, USA 2025
Analytical and Quantitative Light Microscopy Intensive course on quantitative fluorescence microscopy, optics, and image analysis.

TEACHING AND VOLUNTEERING	Teaching Assistant, MC202 Data Structures and MC832 Computer Networks, UNICAMP Techninas - Girls in Tech IEEE Student Branch and Women in Engineering, UEL	2022 2018 – 2020 2016 – 2018
TALKS	DynaTrack, SPIE BiOS 2026 [1] Meso-Regions, IEEE ISBI 2026 [6] Automated reporting of MRI early contrast enhancement in mesothelioma, iMig 2023 Superpixel segmentation: from theory to applications, workshop at SIBGRAPI 2023 [8]	
SKILLS AND INTERESTS	Languages Portuguese (native), English (fluent) Software Python (PyTorch, MONAI, Lightning, scikit-image, napari), OME-Zarr/iohub, Ultrack, ANTs, TensorFlow, PySpark, SQL, C for embedded systems, Git Microscopy Smart microscopy and acquisition control, quantitative phase imaging, virtual staining, light-sheet data	
PUBLICATIONS	[1] T. M. Theodoro, A. Q. Ryan, R. Banks, E. Hirata-Miyasaki, T. Lao, S. Jaisingh, A. Jacobo, S. B. Mehta, and I. Ivanov, “Dynatrack: label-agnostic smart tracking for mapping dynamics of organ development,” in <i>High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond XI</i>, vol. 13862, pp. 23–31, SPIE, 2026. [2] S. Pradeep, A. Imran, Z. Liu, T. M. Theodoro, E. Hirata-Miyasaki, I. Ivanov, M. Bhave, S. Khadka, H. Woosley, C. Arias, and S. B. Mehta, “Contrastive learning of cell state dynamics in response to perturbations,” 2024. [3] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. Ivanov, S. Khadka, M. Bhave, H. Woosley, C. Arias, <i>et al.</i>, “Mapping cell state dynamics with ai-enabled live cell imaging,” in <i>High-Throughput Biophotonics: Imaging, Spectroscopy, and Beyond X</i>, p. PC133300R, SPIE, 2025. [4] E. Hirata-Miyasaki, S. Pradeep, Z. Liu, A. Imran, T. M. Theodoro, I. E. Ivanov, S. Khadka, S.-C. Lee, M. E. Grunberg, H. Woosley, M. Bhave, C. Arias, and S. B. Mehta, “Embedding cell state dynamics via contrastive learning of representations of 3d dynamic imaging datasets,” in <i>NeurIPS 2025 Workshop for Imageomics: Discovering Biological Knowledge from Images Using AI</i>, 2025. [5] A. A. Kalinin, D. Zheng, T. M. Theodoro, I. E. Ivanov, E. Hirata-Miyasaki, S.-C. Lee, Z. Liu, S. Varra, T. Chandler, S. Pradeep, A. Liu, M. Leonetti, A. Arias, B. Huang, and S. Mehta, “Dynacell: An evaluation framework for dynamic 3D virtual staining of live cells,” in <i>The Fortieth Annual Conference on Neural Information Processing Systems Evaluations and Datasets Track</i>, 2026. Poster; accepted. [6] T. M. Theodoro, I. F. Silva, S. Tsim, K. Blyth, and A. X. Falcao, “Meso-regions: 4d segmentation of early contrast enhancement biomarkers for pleural mesothelioma diagnosis,” in <i>2026 IEEE 23rd International Symposium on Biomedical Imaging (ISBI)</i>, pp. 1–5, 2026. [7] T. M. Theodoro, “Towards automatic detection of pleural mesothelioma biomarker in 4d dynamic mr imaging,” master’s thesis, University of Campinas (UNICAMP), Institute of Computing, Campinas, Brazil, 2025. [8] I. Borlido, F. Belém, L. Melo, T. M. Theodoro, I. F. Silva, Z. K. G. do Patrocínio Jr, A. X. Falcao, and S. J. F. Guimaraes, “Superpixel segmentation: from theory to applications,” in <i>(to appear in) 2023 36th SIBGRAPI Conference on Graphics, Patterns and Images (SIBGRAPI)</i>, 2023. [9] S. G. Armato, S. I. Katz, T. Frauenfelder, G. Jayasekera, A. Catino, K. G. Blyth, T. Theodoro, P. Rousset, K. Nackaerts, and I. Opitz, “Imaging in pleural mesothelioma: A review of the 16th international conference of the international mesothelioma interest group,” <i>Lung Cancer</i>, vol. 193, p. 107832, 2024. [10] I. F. da Silva and T. M. Theodoro, “LIDS-UNICAMP: MISE: Multidimensional Image Segmentation Software (initial alpha release),” Aug 2023. [11] T. Theodoro, “Study an Development of an Automation System to Water Treatment by Electrocoagulation Process,” in <i>Congress proceedings.</i>, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2018. [12] T. Theodoro, “Improvement of an Automation System To Water Treatment by Eletrocoagulation Process,” in <i>Congress proceedings.</i>, (Londrina), Annual Congress of Scientific Initiation of UEL, ISSN 2447-4118, 2019.	